Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of a Low-Temperature, Pressure-Integrated in situ Battery Interface Characterization System**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display( ), Secondary Battery (√), Advanced Mobility( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Manufacturing and Measurement/ Evaluation*	*Measuring Instruments*	*Safety Monitoring/Diagnosis Control Instruments*
Concept of Proposed Technology	**Core Concept and Potential Applications** Low-temperature battery operation remains a critical bottleneck for reliable energy storage in electric vehicles, aerospace systems, grid storage, and defense-related applications. Under sub-ambient conditions, battery performance degradation is dominated not by bulk transport limitations, but by interfacial processes such as ion desolvation, interphase evolution, and contact instability [1]. However, most existing low-temperature studies rely on indirect electrochemical measurements and lack direct, materials-resolved insight into interface behavior under realistic operating conditions (see Figure 1)[2].			

The proposed system allows systematic correlation between environmental conditions, interfacial structure, and electrochemical response, enabling mechanistic investigation of ion desolvation, pressure-dependent interphase formation, and interface-dominated kinetics at low temperatures. Through the TIPA global joint R&D program, this platform will establish a new materials-information-driven approach for low-temperature battery research, while expanding the application space of advanced battery diagnostic equipment, as shown in Figure 1.

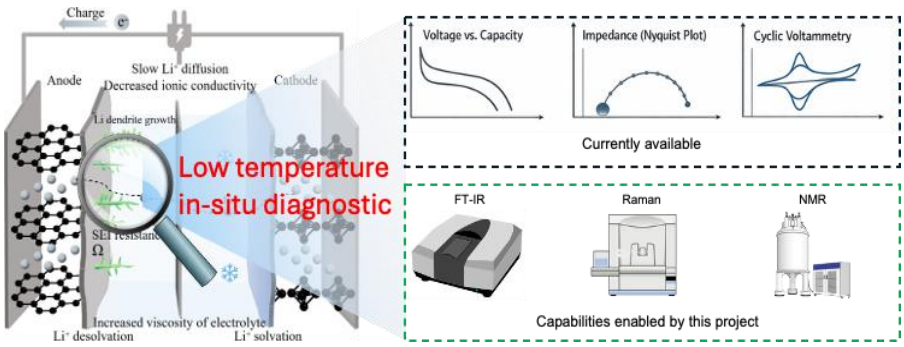

Figure 1. Conceptual comparison between conventional low-temperature battery studies and the proposed low-temperature–pressure integrated in situ battery research platform.

Rationale for Joint R&D Project

Critical Technical Barriers

1. Technical Challenges in Low-Temperature Battery Interface Studies

At low temperatures, battery performance degradation is dominated by interfacial processes rather than bulk ion transport. Ion desolvation, interphase evolution, and metal plating morphology become strongly coupled, leading to increased polarization and unstable electrochemical behavior [1]. Prior studies indicate that interfacial chemistry, electrostatic environment, and mechanical constraint play critical roles. However, these effects evolve concurrently under sub-ambient conditions and cannot be independently resolved using conventional electrochemical measurements. As a result, the mechanisms governing low-temperature interfacial behavior remain poorly understood [3].

2. Limitations of Existing Equipment/Platforms

Current experimental platforms lack the capability to directly link interfacial structure evolution with electrochemical response under coupled low-temperature and mechanical boundary conditions. Most low-temperature systems do not support operando interfacial observation, while existing in situ battery cells are not designed for stable sub-ambient operation or quantitative pressure control [4]. Consequently, the influence of mechanical pressure, despite its known impact on deposition morphology and interphase properties, is often neglected, limiting the relevance of observed interfacial phenomena to realistic operating conditions [4].

Why Global Collaboration?

Within the proposed joint R&D framework, the proposing team contributes core capabilities in interfacial electrochemical analysis, operando characterization, and mechanistic modeling to interpret how interphase chemistry, electrostatic interactions, and ion migration barriers govern low-temperature interfacial behavior. Addressing the above challenges requires the integration of these

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	research capabilities with complementary system-level experimental platform development. Progress in low-temperature battery research further demands advanced experimental infrastructure capable of precise low-temperature control, quantitative pressure regulation, and stable in situ observation of electrochemical interfaces. Such infrastructure is essential to provide the experimental access required for hypothesis-driven investigation and mechanistic interpretation of interface-dominated phenomena. Consequently, the objective of this project is to develop an integrated low-temperature, pressure-controlled in situ battery characterization module that enables direct, materials-resolved investigation of interface-controlled electrochemical phenomena. By treating temperature and pressure as active, independently tunable variables and systematically linking them to interfacial structure evolution and ion transport behavior, the proposed effort establishes a unified experimental and analytical framework that cannot be achieved through isolated development of mechanisms or hardware alone.
Key Words	1. Low temperature battery evaluation system 2. Interface-dominated electrochemical diagnostics 3. Pressure-coupled in-situ observation platform 4. Lithium ion desolvation 5. Multi-scale mechanistic modeling

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	**R&D Objectives** The overall objective of this project is to develop and validate a low-temperature, pressure-integrated in situ battery evaluation framework that bridges advanced interfacial diagnostics with practical testing workflows. By integrating pressure-enabled in situ characterization, benchmarking performance across representative battery chemistries, and establishing standardized, user-oriented evaluation procedures, the project aims to translate advanced experimental capability into a reliable and broadly applicable low-temperature battery evaluation solution. **Detailed Objectives** ● Specific Objective 1: Development of a Commercializable Low-Temperature Evaluation Platform with In Situ Investigation Capabilities This objective aims to develop a low-temperature–pressure coupled in situ battery testing platform that enables stable electrochemical operation and direct access to interfacial structure under sub-ambient conditions. By integrating precise temperature control, quantitative pressure application, and in situ probing within a single system, temperature and pressure are treated as active experimental variables, providing a foundation for systematic investigation of interface-dominated electrochemical behavior. ● Specific Objective 2: Systematic Evaluation of a Low-Temperature Battery Testing Platform and Development of User-Oriented Testing Workflows This objective focuses on evaluating the performance of the low-temperature testing platform across representative battery chemistries. By benchmarking electrochemical behavior under defined temperature and pressure conditions, the project will establish the applicability and operational boundaries of the platform. In parallel, standardized testing workflows and operating procedures will be developed to enable consistent system operation and streamlined data interpretation. ● Specific Objective 3: Development of Engineering-Oriented Low-Temperature Battery Design Guidelines This objective aims to derive practical, engineering-oriented low-temperature battery evaluation guidelines from standardized platform outputs. By correlating system-level metrics, such as polarization response, impedance evolution, and pressure sensitivity, with battery performance, the project will translate evaluation results into actionable guidance for material selection, cell configuration, and operating condition optimization.			

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R&D Target Specifications

1. In Situ Spectroscopic Observation of Lithium-Ion Transport at Low Temperature

   This project employs in situ spectroscopic techniques, including Raman and FTIR spectroscopy, to directly probe lithium-ion transport and interfacial chemical evolution during low-temperature battery operation. By integrating in situ spectroscopy with controlled electrochemical cycling under sub-ambient conditions, the project establishes a direct experimental basis for correlating interfacial transport behavior with electrochemical response.

2. Mechanistic Modeling of Low-Temperature Lithium-Ion Transport Based on Experimental–Theoretical Integration

   This project develops a mechanistic framework by integrating experimental observations with theoretical and multi-scale modeling approaches to capture the relationships among interfacial structure, ion transport pathways, energy barriers, and electrochemical response.The resulting model provides a quantitative basis for understanding interface-dominated transport behavior at low temperatures and supports rational interpretation of experimental results.

3. Development of a Wide-Temperature-Range, Pressure-Coupled In Situ Battery Testing Platform Chamber

   This project aims to develop an in situ battery testing platform capable of stable operation across a wide temperature range while enabling quantitative pressure application. The platform supports simultaneous electrochemical cycling and in situ characterization under controlled thermal and mechanical boundary conditions as a standardized experimental tool for investigating pressure-dependent interfacial lithium-ion transport behavior. Target specifications are summarized in Table 1 and serve as benchmark criteria for system optimization and experimental design.

Parameter	Description	Remark
Temperature Range	*−100 °C to RT*	*Resolution: 0.1 °C*
Typical Average Cooling Rate	*20 °C/min*	—
Maximum Applied Force	*200 N*	—
Force Sensor Resolution	*≤ 0.5 N*	—
Acceptable Electrode Size	*20 mm × 20 mm*	*Max. Pressure: 500 kPa*
Maximum Stacked Cell Thickness	*3 mm*	—
Optical Working Distance	*< 5 mm*	—
Leak Rate	*≤ 1 × 10^{-6} mbar·L·s^{-1}*	*Air equivalent (He leak test)*

Table 1. R&D Target Specifications of the Integrated Low-Temperature In Situ Battery Testing Platform

Consequently, these specifications define the minimum performance thresholds required for the platform to function as a reliable experimental system rather than a proof-of-concept, enabling systematic investigation of coupled temperature–pressure effects on battery interfaces.

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R&D Plan	● Year 1 (6 months): Platform Architecture Definition and In Situ Feasibility Establishment Year 1 focuses on defining the architectural requirements of a pressure-compatible in situ battery characterization platform and establishing the feasibility of interfacial observation under electrochemical operation. Key tasks include: 1. Defining the functional and structural design space for an in situ battery chamber that supports mechanical loading, optical access, and electrochemical diagnostics. 2. Establishing uniform pressure transfer and environmental sealing concepts, together with baseline pressure regulation and monitoring strategies. 3. Developing cutter-enabled sample preparation approaches to enable controlled exposure of interfacial regions for operando spectroscopic interrogation. 4. Identifying key design constraints and interface requirements to guide subsequent low-temperature and pressure integration. ● Year 2 (12 months): Platform Optimization and Initial Low-Temperature–Pressure Integration Year 2 emphasizes integrating low-temperature control with quantitative pressure capability and validating system-level performance under coupled environmental conditions. Key tasks include: 1. Integrating a low-temperature control module into the in situ battery platform based on feasibility outcomes from Year 1. 2. Evaluating the feasibility of in situ Raman and FTIR measurements under electrochemical cycling to assess signal accessibility, spatial resolution, and measurement stability. 3. Implementing mechanically stable co-integration structures that enable simultaneous low-temperature operation and controlled pressure application. 4. Verifying thermal stability, pressure reproducibility, and compatibility with electrochemical cycling under combined temperature–pressure conditions. 5. Using experimental feedback to iteratively refine platform design and operating parameters. ● Year 3 (12 months): Dynamic Pressure Control and Mechanistic Investigation Year 3 focuses on enabling dynamic pressure operation and leveraging the validated platform for mechanistic studies of pressure-sensitive interfacial phenomena. Key tasks include: 1. Developing automated pressure regulation with real-time feedback control to enable static and dynamic pressure modulation during electrochemical cycling. 2. Conducting preliminary benchmarking across representative battery chemistries to evaluate operational stability and measurement feasibility. 3. Systematically mapping stable operating windows across combined temperature–pressure conditions. 4. Investigating pressure-dependent interfacial processes, including lithium-ion transport behavior, interfacial contact evolution, and interphase formation pathways at low temperatures. 5. Integrating experimental observations with theoretical and multi-scale modeling to

	establish a mechanistic framework for lithium-ion transport under coupled temperature–pressure conditions. ● Year 4 (6 months): Model Validation, Platform Verification, and Demonstration Year 4 is dedicated to validating mechanistic understanding and demonstrating platform reliability and readiness for broader application. Key tasks include: 1. Validating the mechanistic transport model through targeted in situ experiments conducted within verified operating envelopes. 2. Comparing model predictions with experimental observations to assess consistency and predictive capability. 3. Verifying platform stability, reproducibility, and scalability under extended operation. 4. Demonstrating the integrated low-temperature, pressure-coupled in situ platform as a reliable experimental tool for interface-focused battery research.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Overall Role: To lead mechanism-driven investigation and experimental validation using the low-temperature, pressure-coupled in situ platform, and to provide actionable feedback that informs platform optimization and data interpretation. 1. Conduct fundamental studies of low-temperature battery interfaces, including ion desolvation, interphase formation, and interface chemistry under coupled temperature–pressure conditions. 2. Perform operando, materials-resolved interfacial characterization using in situ spectroscopic techniques (e.g., Raman, FTIR). 3. Lead data analysis and multi-scale modeling to establish structure-property-performance relationships. 4. Develop and validate mechanistic models describing pressure-sensitive lithium-ion transport behavior. 5. Contribute validated datasets, benchmarking results, and interpretation frameworks to support standardized evaluation and user-oriented deliverables.
Expected R&R of Korean SME Partner	Overall Role: To lead engineering development, system integration, and validation of the low-temperature, pressure-integrated in situ battery characterization platform toward a deployable evaluation system. 1. Design and integrate core hardware modules, including low-temperature control, pressure application, in situ observation, and system control units. 2. Establish reproducible and calibrated temperature and pressure control under sub-ambient operating conditions. 3. Optimize platform architecture and operating parameters based on experimental feedback. 4. Ensure system stability, reproducibility, and reliability during extended coupled temperature–pressure operation. 5. Support platform finalization through standardization of operating protocols, benchmarking procedures, and testing workflows.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Expected Platform and Technology Impact 1. Establishment of a pressure-enabled, low-temperature in situ battery characterization platform that integrates thermal control, quantitative pressure modulation, and operando interfacial diagnostics within a unified system. 2. Expansion of low-temperature and in situ battery evaluation capability to address interface-dominated phenomena that cannot be accessed using conventional pressure-free or bulk testing approaches. 3. Differentiation of low-temperature battery characterization through the use of pressure as an active experimental variable for probing interfacial stability and transport behavior under realistic operating conditions. ● Expected Original Technical Outcomes 1. Identification and validation of interface-sensitive electrochemical and spectroscopic metrics for standardized low-temperature battery evaluation. 2. Direct correlation between interfacial evolution and macroscopic electrochemical response enabled by coupled low-temperature–pressure in situ characterization. 3. Development of a standardized, user-oriented evaluation framework that integrates operando diagnostics with electrochemical testing. 4. Establishment of a validated mechanistic framework for interpreting interface-dominated lithium-ion transport under sub-ambient and pressure-coupled conditions. ● Broader Impact and Adoption Potential 1. Addressing the growing need for advanced battery evaluation tools capable of probing low-temperature and interface-sensitive behavior. 2. Enabling assessment of interfacial stability and transport processes relevant to cold-climate operation and high-reliability applications. 3. Providing a broadly applicable experimental framework for both fundamental studies and applied battery development. 4. Strengthening the research and development ecosystem through standardized benchmarking capability and reproducible evaluation infrastructure.

6. References

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(4) In situ monitoring of lithium-ion battery degradation using an electrochemical model, Applied Energy, 2019, 250, 685–696